

EXP 23: Prolog Program to Implement Family Tree

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Aim:

To develop a Prolog program that represents a family tree using facts and rules, so that relationships such as father, mother, grandfather and sibling can be queried.

Algorithm:

- Step 1: Start the program.
- Step 2: Store information about family members and their genders.
- Step 3: Store parent-child relationships in the database.
- Step 4: Use the stored facts to identify fathers and mothers.
- Step 5: Determine grandfather relationships through parent links.
- Step 6: Identify siblings by checking whether two individuals share a common parent.
- Step 7: Execute the required relationship query.
- Step 8: Display the corresponding family relationship details.
- Step 9: Stop the program.

Source Code:

% Program to implement a Family Tree

```
male(john).
male(michael).
male(david).
male(james).

female(susan).
female(linda).
female(emma).
female(sophia).

parent(john, michael).
parent(john, linda).
parent(susan, michael).
parent(susan, linda).
parent(michael, david).
parent(michael, emma).
parent(linda, james).
parent(linda, sophia).

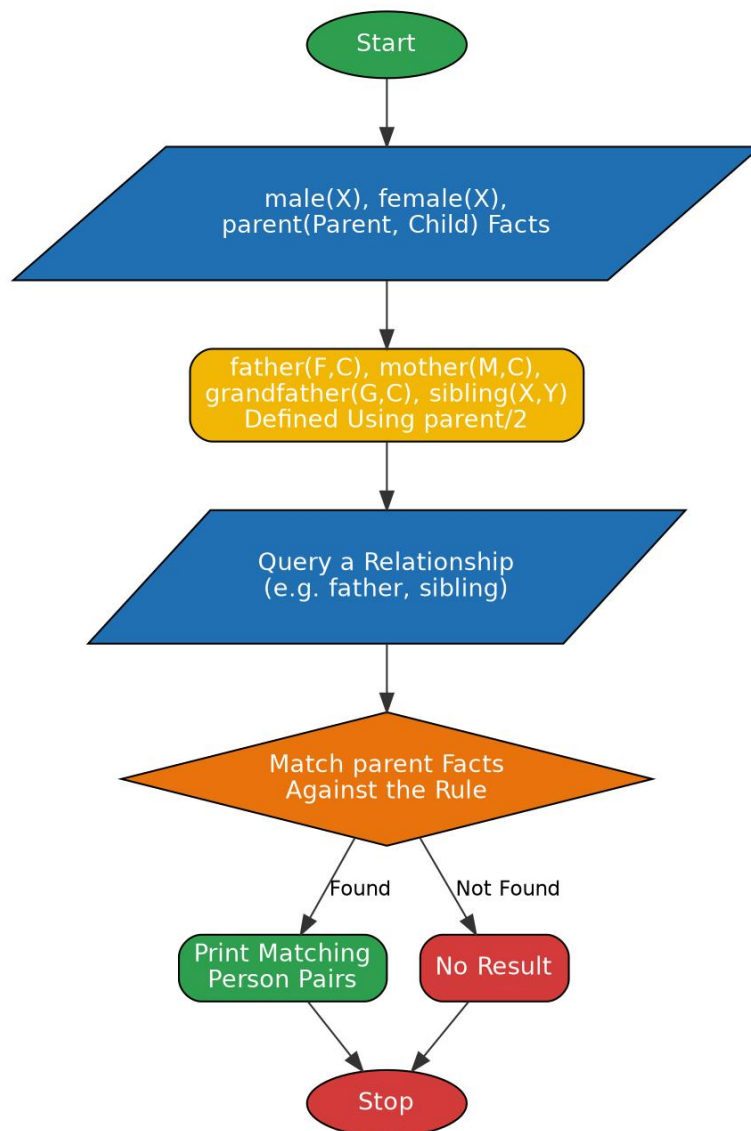
father(F, C) :- male(F), parent(F, C).
mother(M, C) :- female(M), parent(M, C).
grandfather(G, C) :- male(G), parent(G, P), parent(P, C).
sibling(X, Y) :- parent(P, X), parent(P, Y), X \= Y.
```

```

show_fathers :- forall(father(F, C), format("~w is the father of ~w~n", [F, C])).
show_mothers :- forall(mother(M, C), format("~w is the mother of ~w~n", [M, C])).
show_grandfathers :- forall(grandfather(G, C), format("~w is the grandfather of ~w~n", [G, C])).
show_siblings(P) :- setof(X, sibling(P, X), L), forall(member(S, L), format("~w~n", [S])).

```

Flowchart:



Output:

```
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?- cd('C:/Users/yanar/Documents/CSA1717').
true.

?- [family].
true.

?- show_fathers.
john is the father of michael
john is the father of linda
michael is the father of david
michael is the father of emma
true.

?- show_mothers.
susan is the mother of michael
susan is the mother of linda
linda is the mother of james
linda is the mother of sophia
true.

?- show_grandfathers.
john is the grandfather of david
john is the grandfather of emma
john is the grandfather of james
john is the grandfather of sophia
true.

?- show_siblings(michael).
linda
true.

?- show_siblings(linda).
michael
true.
```

Result:

Thus the Prolog program to implement the family tree was written, executed successfully and the output was verified.